

Investigating Explainable Machine Learning Models for Dynamic Credit Scoring Using Behavioural Financial Data

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1. Introduction

Credit scoring is a fundamental mechanism used by financial institutions to evaluate borrower's creditworthiness, manage lending decisions, and manage credit risk (Rajender, et al., 2025). In Malaysia, credit assessment commonly relies on credit reporting agencies such as CCRIS and CTOS, which aggregate historical financing and repayment records from formal financial institutions (Bank Negara Malaysia, n.d.). While these systems have served as foundational tools for credit risk assessment, they are limited in their ability to reflect a borrower's financial evolves over time. At the same time, the rapid growth of digital financial services such as online banking, e-wallets and mobile payment platforms has generated large volumes of daily behavioural financial data that existing credit scoring systems do not systematically utilise. Furthermore, as machine learning models become increasingly adopted to improve credit risk prediction, many operate as opaque black-box systems that provide limited explanation regarding how credit decisions are generated. This raises significant ethical and regulatory concerns as every decision cannot be explained to regulators and borrowers. These three interconnected limitations, which are static data reliance, underutilised behavioural signals, and lack of model transparency represent a critical gap in contemporary credit scoring practice. This paper is organised as follows: Section 2 presents the literature review across three research domains and a similar systems comparison, Section 3 describes the research methodology, and Section 4 concludes.

2. Literature Review

2.1 Traditional Credit Scoring and Its Limitation

Credit scoring has served as the standard mechanism for evaluating borrower creditworthiness and managing lending risk across financial institutions. Conventional approaches apply standardised scoring models to generate a numerical risk rating that guides lending decisions using a structured credit bureau data such as repayment history, outstanding debt obligations, income stability, and employment status (Malik et al., 2025). These conventional models have been widely adopted due to their operational simplicity and regulatory acceptance to provide consistent and auditable assessments across large borrower populations (Malik et al, 2025).

In Malaysia, credit assessment is supported by two principal credit reporting systems, which are CCRIS and CTOS. CCRIS, managed by Bank Negara Malaysia, collects financing and repayment credit records from financial institutions and is updated monthly (CTOS, 2026). On the other hand, CTOS is an approved credit reporting agency that supplements CCRIS data with legal records, trade references, and bankruptcy information for a more comprehensive credit profile. Together, these systems form the foundational infrastructure of credit risk assessment in Malaysia and have supported structured lending decisions within the formal financial sector.

Although these systems remain useful, some limitations of traditional credit scoring have become increasingly noticeable as financial activities has shifted towards digital platforms, and the volume of alternative financial data has grown substantially. One of the limitations is traditional credit scoring mainly provides a historical view of a borrower's financial position. Even though CCRIS refreshes monthly, each update only reflects the same narrow categories of formal credit obligations

such as loan repayments, credit card balances, and outstanding financing. This means that behavioural signals occurring between formal reporting cycles remain entirely invisible to the system (Malik, Kaul, Gautham Jagthap, Harshit Jamley, & Chopra, 2025). For example, a borrower who experiences a sudden income drop or begins managing finances more responsibly through digital banking may see these changes only after a later credit event, such as a delayed or missed repayment.

The second limitation is the systematic exclusion of thin-file borrowers. Thin-file borrowers are individuals who have little or no formal credit history. These individuals will appear invisible to credit bureaus because the system rely heavily on formal borrowing and repayment records. Even though they may actively use digital financial services and manage their finances responsibly, they may face difficulties when applying for a credit. The World Bank Group (2025) reported that approximately 1.4 billion adults worldwide remain financially underserved due to limited formal credit histories. This represents a structural exclusion embedded in traditional credit assessment models where current system disproportionately disadvantages economically active individuals in emerging digital economies. In Malaysia, digital banking penetration exceeded 90 percent of the adult population by 2023. This means that the volume of behavioural financial data being generated through digital platforms has grown substantially while the structure of the formal credit reporting mechanisms remained unchanged (Bank Negara Malaysia).

The last limitation is the static architecture of the risk assessment. Traditional scoring models normally produce a risk score based on information available at a particular point in time, so the score may be slow to reflect changes in a borrower's financial circumstances, such as changes in income, spending patterns, repayment behaviour, or financial commitments (Verma & Kumar, 2024). As a result, financial institutions may be slower to identify increasing or decreasing borrower risk when those changes first appear through recent financial behaviour.

Overall, traditional credit scoring remains an important foundation for credit assessment because it provides structured and reliable historical information. However, its reliance on formal credit records and fixed-point assessments may limit its ability to reflect recent behavioural changes and assess thin-file borrowers. Therefore, complementary approaches are needed to analyse behavioural financial data and support more responsive, dynamic, and inclusive credit risk assessment.

2.2 Behavioural Financial Data and Machine Learning for Dynamic Credit Scoring

In today digital world, financial services such as online banking, e-wallets, mobile payment platforms, and buy-now-pay-later services is rapidly growing and has generated large volumes of daily behavioural financial data (Malik, Kaul, Gautham Jagthap, Harshit Jamley, & Chopra, 2025). Unlike formal credit records that only capture periodic information about existing credit obligations, behavioural data can reveal more recent patterns in how individuals spend, repay, transact, save, and manage their financial commitments. This pattern may provide more current and detailed view of a borrower's financial condition than bureau data alone can provide (Verma & Kumar, 2024; Malik et al., 2025).

Previous studies suggest that transactional and behavioural data can strengthen credit risk assessment by allowing changes in financial behaviour to be identified earlier. For example, Verma and Kumar (2024) demonstrated that incorporating behavioural transaction data into credit risk models improved classification accuracy by 3 percent compared to models relying fully on traditional bureau data. Roland Abi (2025) further found that machine learning models trained on behavioural and transactional data can generate early warning signals up to 30 to 90 days before a borrower misses a formal payment, a capability structurally impossible for static bureau-based systems to replicate. This is particularly important because individual financial behaviour is not fixed but changes in response to income fluctuations, employment changes, spending habits, and personal circumstances.

Therefore, a scoring approach that considers recent behavioural information may provide a more responsive assessment of changing borrower risk (Malik, Kaul, Gautham Jagthap, Harshit Jamley, & Chopra, 2025).

Based on the literature reviewed, there are four behavioural financial indicators that have been consistently identified as measurable inputs for dynamic credit scoring, which are spending behaviour, repayment behaviour, transaction frequency, and financial management behaviour. Firstly, **spending behaviour**. This reflects how a borrower manages outgoing financial resources: irregular or excessive spending relative to income may signal financial stress, while stable, proportionate spending suggests greater financial responsibility (Verma & Kumar, 2024).

Second behavioural indicator is the **repayment behaviour**. Repayment behaviour is one of the most direct indicators of payment discipline and default risk: borrowers who pay instalments, bills, and other commitments on time are generally more reliable, while repeated late or missed payments may indicate higher credit risk (Rajender, et al., 2025; Abi, 2025).

The third indicator is **transaction frequency**, which reflects a borrower's level of financial engagement and account activity. Regular transaction patterns indicate active financial management, while highly irregular activity may reduce the predictability of a borrower's financial behaviour (Malik, Kaul, Gautham Jagthap, Harshit Jamley, & Chopra, 2025). This is particularly valuable for thin-file borrowers who lack formal credit histories but generate large digital transaction records daily.

Finally, the fourth indicator is the **financial management behaviour**. This captures how borrowers handle broader financial responsibilities including savings discipline, debt management, and budgeting consistency (Verma & Kumar, 2024; Jayanthi & Pruseth, 2026). Unlike the previous three, this indicator reflects a borrower's overall financial discipline and commitment management habits.

Overall, these behavioural indicators are broad categories rather than direct model inputs. Therefore, raw financial records must first be transformed through feature engineering into measurable variables that can be used as inputs for machine learning models. However, as behavioural financial data is large in volume, diverse, and continually evolving, the transformation process introduces analytical challenges that can be difficult to address using conventional statistical approaches alone (Verma & Kumar, 2024). **Machine learning** is therefore considered because it is suitable to identify complex non-linear relationships and detect hidden patterns within large and high-dimensional datasets (Jayanthi & Pruseth, 2026).

Three machine learning techniques have been identified in the literature as particularly suitable for analysing behavioural financial data in dynamic credit scoring, which are Random Forest, XGBoost, and Artificial Neural Networks (ANN). Random Forest is an ensemble method that handles non-linear relationships robustly and resists noisy behavioural data, characteristics important given the diversity of digital financial signals (Miracle, Shola, & Seun, 2024). Random Forest can also produce feature importance scores that provide initial insight into which behavioural indicators contribute most to predictions, offering a degree of interpretability not available in deep learning models (Abi, 2025). XGBoost has demonstrated consistently strong predictive performance on structured tabular financial data and is widely validated in credit risk prediction literature (Sampathkumar, Pathak, Ramaraj, Kotha, & Patel, 2026; Jayanthi & Pruseth, 2026). Lastly, Artificial Neural Networks can model highly complex non-linear relationships within large datasets, but their internal decision process is more difficult to interpret. (Abi, 2025; Malik, Kaul, Gautham Jagthap, Harshit Jamley, & Chopra, 2025)

Overall, behavioural financial data provides more recent information about borrower activity, while machine learning offers the analytical capability to identify complex patterns within that data. However, strong predictive capability alone is not sufficient for

credit scoring because financial institutions and borrowers must also understand how a prediction is produced. Therefore, this creates the need for Explainable Artificial Intelligence.

2.3 Explainable Artificial Intelligence for Transparent Credit Scoring

Credit scoring decisions directly affect individuals' access to financial services such as loan approvals, credit limits, and interest rates. This makes transparency and accountability fundamental requirement of any automated credit assessment system rather than just a desirable feature (Izah et al., 2025). In highly regulated domains like financial services, stakeholders such as regulators, financial institutions, and borrowers need to understand how automated predictions are generated rather than just know them as model that produces a number (Anang, et al., 2024).

When machine learning models operate as black-box systems, this requirement breaks down on two sides. Firstly, financial institutions face difficulties justifying automated credit decisions to regulators and customers, and secondly, borrowers who receive unfavourable credit decisions are unable to understand which aspects of their financial behaviour contributed to the outcome or how they might improve their creditworthiness (Izah et al., 2025; Rajender, et al., 2025). This is supported by Anang et al. (2024), who found that 78 percent of fintech professionals surveyed considered model explainability essential for regulatory compliance. However, the same study also found that feature-level explanations remain limited in many existing credit scoring systems.

Moreover, In August 2024, the European Union AI Act formally classifies credit scoring as a high-risk AI application and requires transparent documentation, individual explanations, and human oversight for every decision made (Anang, et al., 2024). Furthermore, in Malaysia, Bank Negara Malaysia's responsible AI discussion paper (2022) and the Financial Sector Blueprint 2022 to 2026 signal the same direction. This regulatory trajectory makes XAI not just

academically relevant, but practically necessary.

Explainable AI (XAI) refers to techniques designed to make machine learning outputs interpretable while preserving the model's predictive performance (Ravi et al., 2025). SHAP and LIME are widely identified in recent literature as particularly applicable to credit scoring. SHapley Additive exPlanations (SHAP) is grounded in game theory and assigns each input feature a mathematically justified contribution value for a specific prediction (Ravi et al., 2025). SHAP works at two levels, which are local and global. For local, it can explain why a particular borrower received a particular score, while at global, it can show which specific features matter the most across all predictions (Rajender, et al., 2025). For example, SHAP could tell a borrower that their repayment consistency added a specific point to their score while their spending volatility reduced it by another, giving specific information rather than a vague category label (Ravi et al., 2025). On the other hand, Local Interpretable Model-agnostic Explanations (LIME) explains specific predictions by constructing a simpler interpretable model built around that prediction to approximate how the complex ML model behaved for that case (Abi, 2025). Moreover, LIME is model-agnostic, meaning it can work with any of the ML models proposed in this framework without modification.

Despite the demonstrated capability of XAI techniques in recent studies, their integration into production credit scoring systems remains limited. Current fintech credit scoring implementations that is used by platform-based lenders and digital banks typically provide surface-level explanations only. For example, they present broad category labels such as "repayment history" or "credit utilisation" without quantifying how much each factor contributed to a specific decision or in which direction (Anang, et al., 2024; Rajender, et al., 2025). This limits the practical utility of such explanations for both accountability and financial improvement, since borrowers cannot identify which behaviour to address or by how much.

Collectively, these three domains reveal that traditional credit scoring, behavioural financial data, and explainable AI have each been examined individually, but limited research integrates all three within a unified framework, as the following similar systems analysis demonstrates.

2.4 Similar System

The comparison of five similar systems is presented in Table 1, where each system is evaluated based on its main approach, findings, and limitations relative to the present research.

As summarised in Table 1, each system contributes to at least one element of behavioural data, dynamic scoring, machine learning, or explainability, but none integrate all four within a single framework, and several rely on static datasets that do not support continuous behavioural updates.

Table 1 Similar system comparisons with research gaps identified

Journal / Conference (Year)	Authors	Title	Key Findings	Research Gaps
World Journal of Advanced Research and Reviews (2025)	Abi, R.	Machine Learning for Credit Scoring and Loan Default Prediction Using Behavioral and Transactional Financial Data	The integration of behavioural signals, such as spending patterns, payment timing, and mobile usage, with machine learning improves default prediction and may support financial inclusion in higher-risk environments.	Although the study discusses XAI techniques such as SHAP and LIME, it does not provide an integrated framework that addresses temporal data misalignment or dynamic feedback loops.
International Journal of Science, Engineering and Technology (2025)	Malik, P., Kaul, V., Jagthap, G., Jamley, H., and Chopra, H.	Dynamic Credit Scoring with Real-Time Transactional and Behavioral Data Using Deep	The study formulates dynamic scoring as a Markov Decision Process and applies Deep Reinforcement	The study acknowledges the black-box nature of Deep Reinforcement Learning and identifies

		Reinforcement Learning	Learning to update creditworthiness in real time, supporting continuous risk estimation and adaptive credit limit adjustment.	SHAP-based decomposition as a future direction, but it does not include a dedicated and validated XAI layer for real-time stakeholder explanations.
2025 7th International Conference on Information Systems and Computer Networks — ISCON (2025)	Rajender, C., Gururani, S., Mamadiyarov, Z., Toumashova, M., Khorolskaya, T., and Naresh, R. B.	Explainable AI for Credit Scoring in Cloud Enterprise Banking Systems	A hybrid XAI approach combining LIME, SHAP, and rule-based aggregation achieved 89.2% accuracy and improved interpretability scores by 12% compared with traditional baseline models.	The framework was tested using a static historical dataset of 50,000 applications and does not address dynamic, high-frequency behavioural data or real-time score updates.
Journal of Advanced Computing Systems (2026)	Luo, Z., and Yu, M.	An Empirical Comparison of Feature Engineering Strategies from Non-Traditional Data for Thin-File Borrower Credit Assessment	Behavioural features derived from payment patterns produced the highest marginal predictive improvement for thin-file borrowers and reduced the performance gap compared with thick-file borrowers by 49.7%.	The study uses a static evaluation design and does not examine concept drift or how the importance of behavioural features may change over time as economic conditions evolve.
International Journal for Multidisciplinary Research (2026)	Jayanthi, M. K., and Pruseth, S.	An Explainable Machine Learning Framework for Dynamic Credit Portfolio Risk Assessment	The study integrates an optimised Random Forest model with SHAP-based local and global explanations	The framework focuses mainly on portfolio-level risk and traditional financial indicators

		nt and Default Prediction Using Random Forest Optimisation	ns, reducing default misclassification by 23% and supporting explainable credit assessment	. It does not explicitly use high-frequency behavioural transaction data as the main input for individual dynamic credit scores.
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3. Methodology

3.1 Target user

The proposed explainable dynamic credit scoring framework is designed to serve three primary user groups within the financial services ecosystem.

The first user group is financial institutions, including commercial banks, digital banks, and fintech lenders, whose risk management teams and credit officers would use the framework to generate dynamic credit score predictions based on borrower behavioural financial data.

The second user group is borrowers. This includes individuals who apply for credit products including personal loans, credit cards, and buy-now-pay-later facilities. Borrowers may benefit from the framework through XAI-generated explanations that accompany credit decisions. This feature enables them to understand which specific financial behaviours contributed positively or negatively to their predicted credit score and what corrective actions are available to improve their creditworthiness over time.

The third user group is credit analysts and data professionals responsible for interpreting model outputs and supporting lending decisions. The framework's decision support layer provides this group with both the predicted credit score and its behavioural explanation simultaneously, enabling more informed, transparent, and auditable lending decisions than black-box ML systems currently permit.

3.2 Sampling method

To investigate the proposed framework, primary data will be collected through a structured questionnaire survey. This study targets two groups of respondents whose perspectives together may validate the proposed framework at both user and institutional level. The total target sample size of this research is 150 respondents, which consists of 120 respondents from the first group, and 30 respondents for the second group.

The first group targeted financial service users who aged above 18, have experience with credit-related financial services and are actively use digital financial services such as online banking, e-wallets, credit cards, buy-now-pay-later platforms, or mobile payment applications. This group represents the borrower perspective as their perceptions best reflect real creditworthiness. Their views on transparency in credit decisions are also directly relevant to validating the explainability requirements of the proposed framework.

The second group includes financial or fintech professionals with direct experience in credit assessment and financial operations, such as banking staff, credit officers, fintech employees, data analysts, and risk management professionals. This group provides institutional and technical perspectives on the practical applicability of the proposed behavioural indicators and the operational feasibility of explainable AI techniques.

Stratified sampling is selected as the primary sampling strategy because the study population consists of two distinct subgroups that each contribute different knowledge levels, institutional access, and relevance to the research objectives. Therefore, stratification ensures that both user and professional perspectives are represented at planned proportions while strengthening the representativeness and credibility of the collected data. Within each stratum, respondents will be recruited through convenience sampling due to practical access constraints — online channels and social media for the user group, and professional

networks such as LinkedIn and finance-related academic contacts for the professional group.

3.3 Data collection method

A quantitative research approach will be used through a structured questionnaire. This method is suitable for the study because it allows measurable information to be collected from a relatively large number of respondents regarding behavioural financial indicators, dynamic credit scoring, and perceptions towards transparency in automated credit decisions.

The questionnaire will be distributed through an online survey platform and divided into four sections. The first section will collect demographic information such as age, employment status, income level, and experience with digital financial services. The second section will collect respondents' views on behavioural financial indicators, including spending behaviour, repayment behaviour, transaction frequency, and financial management behaviour. The third section will examine respondents' understanding and acceptance of dynamic credit scoring, including whether credit scores should be updated when financial behaviour changes. The final section will collect participants' opinions on explainable AI, such as whether borrowers should receive clear explanations about the factors influencing their credit scores.

For industry professionals, additional questionnaire items will be included to collect more detailed information. For example, their views will be obtained regarding the suitability of behavioural data for credit assessment, the practical use of machine learning and XAI techniques, and the possible challenges of implementing explainable dynamic credit scoring within financial institutions.

4. Conclusion

In conclusion, this study focused on two main objectives. Firstly, it examined the limitations of traditional credit scoring systems that mainly rely on historical credit records and static risk assessments. Other than that, it proposed an integrated conceptual framework that combines behavioural financial data,

machine learning models, and explainable artificial intelligence to support dynamic credit scoring. Previous research shows that borrower financial behaviour can change over time due to income fluctuations, spending patterns, repayment habits, transaction activity, and financial management practices. However, these changes may not be reflected immediately in traditional credit records because existing credit scoring systems mainly depend on formal information such as loan repayments, outstanding debt, and credit history. As a result, financial institutions may not be able to identify changes in borrower risk at an early stage, while thin-file borrowers may also receive limited assessment due to the lack of formal credit records. Through the proposed framework, four behavioural indicators are included in the credit assessment process, namely spending behaviour, repayment behaviour, transaction frequency, and financial management behaviour. These indicators are analysed using machine learning models to generate updated credit risk predictions, while XAI techniques such as SHAP and LIME are used to explain how each behavioural factor influences the prediction. The comparison of similar systems further shows that although behavioural data, dynamic scoring, machine learning, and explainability have been explored in previous studies, limited research integrates all of these components within one framework for explainable dynamic credit scoring. By establishing the conceptual foundation for this framework, this study provides a useful basis for further empirical research, model evaluation, framework refinement, and the development of more transparent and adaptive credit assessment systems.

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